A03 Bridging Neuroscience and Artificial Intelligence

Course: ITAI-4374

Submitter: Oscar Cortez

Group: Magnolia

February 2, 2026

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1. Neural Network Comparison Analysis

1.1 Biological-to-Artificial Neuron Mapping

Figure 1 (Insert Wolfram Alpha screenshot): "Human brain anatomy" visualization from Wolfram Alpha.
Caption: Figure 1. Wolfram Alpha anatomy visualization used to explore human brain structures.

Figure 2 (Insert Wolfram Alpha screenshot): "Neuron anatomy" diagram from Wolfram Alpha.
Caption: Figure 2. Neuron anatomy diagram showing dendrites, soma, axon, and terminals.

Component Mapping Table

Biological component	Role in biology	Artificial analogue (ANN)
Cell body (soma)	Integrates inputs; triggers firing decision	Processing unit (weighted sum + activation)
Dendrites	Receive synaptic signals from many neurons	Input connections / incoming features
Axon	Transmits action potentials to targets	Output connection (unit output to next layer)
Synapses	Connection strength; plastic changes with experience	Weights (learnable parameters)

1.2 150-Word Neuron Comparison (150 words)

Biological neurons are living electrochemical cells that integrate many synaptic inputs on dendrites, transform that activity in the soma, and send action potentials down an axon to other cells. Artificial neurons copy the same "many inputs → combine → one output" pattern: inputs are multiplied by weights (synaptic strength), summed in a unit (soma), passed through an activation function, and forwarded as an output (axon). The differences are implementation and efficiency. Biology uses spikes, neurotransmitters, and complex local plasticity (e.g., STDP) to learn continuously, while ANNs usually use real-valued activations and global backpropagation on GPUs. Biological networks are massively parallel and energy-frugal, but noisy and hard to program. Artificial networks are precise, fast to train on huge datasets, and easy to deploy, but can be data-hungry, brittle out of distribution, and compute-intensive. This inspiration explains layered feature extraction in CNNs and recurrent loops in RNNs, yet biology still remains far richer.

1.3 Comparative Analysis Table

Biological neural network characteristic	Corresponding ANN feature	Example in modern AI
Hierarchical feature extraction (sensory streams)	Layered representations	CNNs: early edges → mid textures → high-level objects
Recurrent loops and feedback pathways	Recurrence / state	RNNs / LSTMs: temporal state for sequences
Sparse, event-driven signaling (spikes)	Sparse activations / event-driven compute	SNNs, sparse transformers, conditional computation
Local plasticity + modulation (STDP, neuromodulators)	Learning rules + adaptive optimizers	Hebbian/contrastive rules; Adam; meta-learning
Specialized modules (vision, motor, reward)	Modular architectures	Multimodal models; mixture-of-experts; RL agents
Memory systems + consolidation	External memory / retrieval	RAG, differentiable memory, experience replay

2. Brain Architecture and AI Design

2.1 Hierarchical Organization and Specialization

The human brain is organized hierarchically: brainstem (basic life support), subcortical structures (routing, reward, emotion, memory), and cortex (high-level perception, planning, and language). Within cortex, regions specialize—visual areas process vision, auditory cortex processes sound, motor areas control movement, and association areas integrate information for decision making. This specialization shows up in modern AI as separate modules (or subnetworks) that are optimized for particular data types and tasks.

Figure 3 (Insert image): Brain lobes diagram (frontal, parietal, temporal, occipital).
Caption: Figure 3. Major cortical lobes shown for functional reference.

2.2 Mapping Brain Systems to CNNs, LSTMs, and Attention

Vision → CNNs: The visual system processes information in stages (from simple edges to complex patterns). CNNs mirror this with early convolutional filters capturing edges and later layers capturing object parts and semantics.

Memory → LSTMs / external memory: The hippocampal system supports rapid formation of episodic memories and later consolidation. LSTMs provide gated state to maintain and update information over time, and modern agents add explicit memory stores (retrieval and replay) to reduce forgetting.

Attention → Transformers: Biological attention uses top-down control (frontoparietal networks) to prioritize relevant signals. Transformer attention implements a computational analogue by weighting which tokens/features should influence the current representation.

Figure 4 (Insert image): Simplified brain lobe diagram (optional).
Caption: Figure 4. Simplified lobe diagram for quick reference.

2.3 Amygdala, Emotion Processing, and AI Implications

The amygdala is a key hub for emotion processing, especially fear learning, threat detection, and emotionally tagged memory. For AI, "emotion" can be framed as fast affective signals that bias learning and action selection.

- **AI decision making:** add rapid "risk flags" that shift a policy toward safer actions when uncertain or threat-like patterns appear.
- **Risk assessment:** treat rare-but-high-cost outcomes as first-class signals (like a learned hazard prior), not just average reward.
- **Pattern recognition:** allocate more compute to salient cues (novelty, surprise, or danger) via adaptive attention.
- **Adaptive learning:** modulate learning rates or exploration when strong prediction errors occur (neuromodulation-inspired).

3. Integration Analysis and Future Implications (250 words)

Studying biological neural networks suggests design ideas beyond today's backprop-trained, synchronous layers. Brains learn online with sparse spikes, local credit assignment, and strong inductive biases (hierarchical sensory streams, memory systems, and attention). Future AI architectures could combine transformer-style global attention with hippocampus-like episodic memory modules, plus neuromodulatory "gating" signals that reshape learning rates during surprise or threat. Biology also offers solutions to stability–plasticity: synapses change continually without catastrophic forgetting, hinting at replay, consolidation, and structural sparsity as default features rather than add-ons. Critically, biological systems trade raw arithmetic precision for efficiency. The brain runs on ~20 W, using event-driven communication and massively parallel, mixed analog–digital computation, while modern deep networks often require data centers and energy-hungry training. Learning differs too: animals generalize from few examples by leveraging prior structure and self-supervised prediction, whereas many models still need large labeled datasets. Adaptability is another gap: brains reconfigure under damage and shifting goals; most deployed models are fixed after training and fail under distribution shift. Looking forward, biological insights may help solve current limitations in robustness, continual learning, and sample efficiency. Promising directions include spiking neural networks on neuromorphic hardware, predictive-coding objectives for perception, and hybrid agents that couple perception (CNN/transformers) with model-based planning and emotion-inspired risk signals (amygdala-like threat tagging) for safer decision making. Integration between biological and artificial systems could also be literal: brain–computer interfaces for closed-loop rehabilitation and prosthetics. In research, connectomics and single-cell recordings may inspire graph-based architectures that respect wiring constraints and modularity.

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